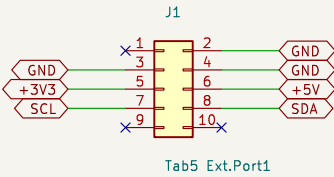
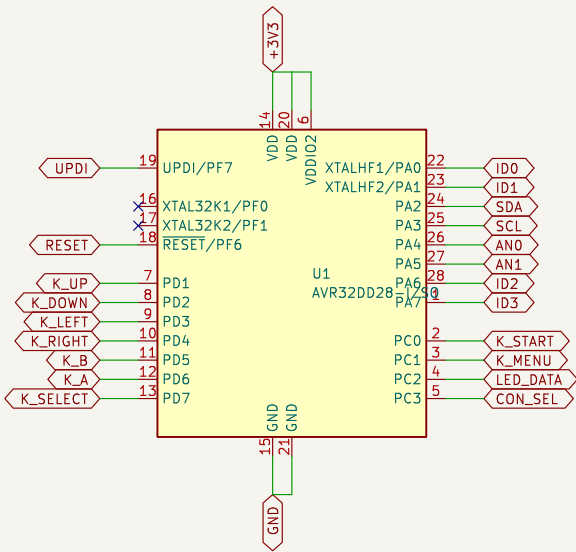
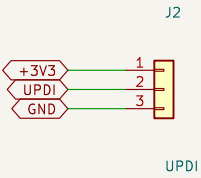


MCU – AVR32DD28 (SOIC–28) running firmware_avrdd (I2C client 0x6D, RetroPad Ext.Port1) – verify pin 1 orientation before fab Buttons – one MCU pin each, switch to GND (internal pull–up); slot table in firmware_avrdd/pinmap.h

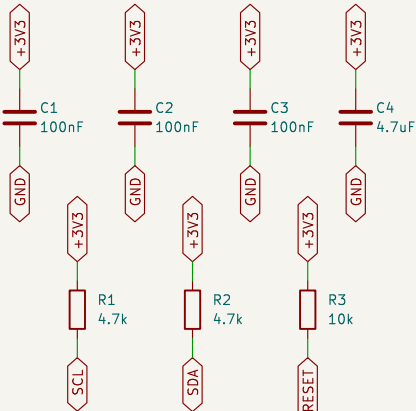
PD1–PD7, PC0–PC3, PF0–PF1: one button per pin (internal pull–ups)



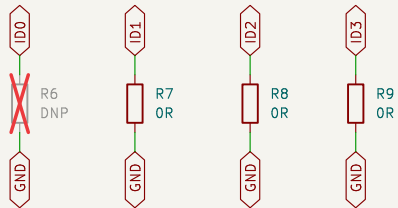
UPDI programming (SerialUPDI / MPLAB Snap)



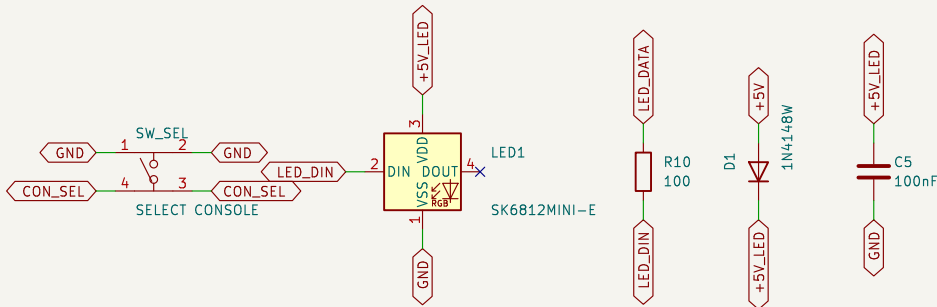
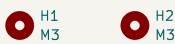
Decoupling (VDD x2, VDDIO2), I2C pull–ups, RESET pull–up



Console ID straps: 0R to GND = bit set. NES / Game Boy / Master System: straps = ID 14 (fit R7, R8, R9), console picked with SW_SEL (NES / GB / SMS)



Mounting holes (M3, 96 mm apart)



Console select: SW_SEL steps NES / GB / SMS (PC3); LED1 shows it (PC2 via R10). LED1 runs from J1's 5 V through D1 (~4.3 V)

